International Islamic University Chittagong

Dept. of Computer Science & Engineering (CSE)

B.Sc. in CSE, Semester Mid-Term Examination, Autumn 2025

Course Code: CSE 4875 **Title:** Pattern Recognition and image processing

Total Marks: 30 **Time:** 1.5 hours

		Marks	CO	DL
1(a)	"One picture is worth more than ten thousand words" – Explain the fields of image processing used in the statement.	2	CO1	C2
1(b)	Why Sampling and Quantization are important in the process of digital image processing?	2	CO1	C2
1(c)	The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes. Explain why we need to process images in low, mid and high level processing with proper example.	3	CO1	C2
1(d)	"A digital image is a representation of a two-dimensional image as a finite set of digital values" – do you agree with the statement? Explain the answer with mathematical formula.	3	CO1	C2

		Marks	CO	DL
2(a)	Justify the statement, "Applying Low-pass filter on an image result in a blurrier image".	2	CO1	C2
2(b)	A 5x4 3 bits/pixel original image is given in ***Fig. 1***.	6	CO2	C3

0	4	7	0
2	2	1	4
5	3	2	6
5	2	2	6
8	8	2	5

Fig. 1

-1	-2	-1
0	0	0
1	2	1

Fig. 2

i. Apply histogram equalization to the image by rounding the resulting image pixels to integer.

ii. Sketch the histograms of the original image and the histogram-equalized image.

iii. Why histogram equalization not produce a perfectly flat histogram?

		Marks	CO	DL
2(c)	A 3x3 mask is given in ***Fig. 2***. Apply this mask on pixel (3, 2) = 3, of image in ***Fig. 1*** and compute the convolution output value for this pixel.	2	CO1	C2

OR

		Marks	CO	DL
2(a)	Monotonically increasing and strictly monotonically increasing function have some specific property. Why these properties are important for histogram equalization?	2	CO1	C2
2(b)	A 5x4 3 bits/pixel original image is given in ***Fig. 1***. i. Sketch the histograms of the original image. Use histogram specification to generate histogram of enhanced image.	6	CO2	C3
2(c)	"When automatic enhancement is desired, equalization is a good approach". Explain the statement with proper justification.	2	CO1	C2

		Marks	CO	DL
3(a)	"Median filter technique is the best way to de-noise the image" – Justify the statement with your own word with proper example.	2	CO1	C3
3(b)	Explain how do human beings perceive color? At HSI color model, how actual and purity of the color are measured? Give the answer with proper diagram?	3	CO2	C2
3(c)	Thresholding is a basic step to convert a gray scaled image into a binary image using a specific threshold value. To find this threshold value is the most critical part. Assume that we have designed a basic algorithm to calculate the average threshold ***T*** (mean gray level value). Express the threshold value in functional form. Now apply the algorithm to the ***Fig. 1*** to convert the gray scale image to binary image.	5	CO2	C3

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Autumn 2025
Course Code: CSE-4805 **Course Title:** Social, Professional and Ethical Issues in Computing
Total marks: 30 *Time: 90 minutes*
[Answer all the questions; in some questions, there might be options; Figures in the right-hand margin indicate full marks.]

1. a)	Technological change often creates new possibilities while introducing unexpected challenges. How can society balance rapid technological development with the need to preserve ethical values, privacy, and human connection?	**CO1**	**U**	*5*
b)	How has the development of modern computing led to the growth of Artificial Intelligence (AI)? Discuss how improvements in computer hardware and software have helped AI change important areas like education, healthcare, and farming. Also, mention some social and ethical challenges that come with these changes	**CO2**	**U**	*5*
	OR			
	Suppose Mr. Y got a job opportunity and joined the job. But the job isn't permanent based on appointment. Then he got an offer from another corporate job with a higher salary. So, Mr. Y left his current job and joined a corporate job. i. Determine the act of Mr. Y based on teleological theories. Briefly explain it. ii. Determine his action based on legality and ethicality. iii. What are the criticisms of this course of action by Mr. Y?			*5*
2.	As technology and social media rapidly evolve, platforms now give everyone the power to share information instantly with millions of people. Sometimes, these tools are used to express opinions, but they can also spread harmful or hateful content. Imagine a situation where a popular influencer posts messages that some users consider offensive or discriminatory, while others defend the posts as expressions of personal opinion. Platform administrators must decide whether to remove the content or allow it to stay online.			
a)	Define "freedom of speech" and "offensive speech" in the context of digital communication, giving one real-life example of each.	**CO1**	**A**	*5*
b)	What ethical and legal factors should guide technology companies when deciding whether to remove harmful content or uphold freedom of expression?	**CO2**	**An**	*5*
	OR	**CO2**	**C**	*5*
	Explain the statement "New Technology, New Risks."How should privacy principles be redefined to effectively address these new risks while ensuring a balance between technological advancement and the protection of individual rights in Bangladesh?			
3. a)	Critically analyze how the ICT Act 2006 of Bangladesh balances the protection of digital rights and freedom of expression with the need to ensure cybersecurity and prevent cybercrimes. Do you think the Act achieves an appropriate balance between these objectives? Justify your answer with examples from the law	**CO2**	**U**	*5*
b)	Explain the key elements of the ICT Act 2006 and the Cyber Security Act 2023. How do these laws differ in addressing digital crimes and ensuring online safety? Discuss the major changes introduced in the new Cyber Safety Ordinance 2025 and their impact on freedom of expression and privacy in Bangladesh.	*CO3*	*A*	*5*

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. Engineering in CSE

Midterm Examination, Autumn 2025

Course Code: **MGT-3601** Course Title: **Industrial Management**

Time: 1 hour 30 minutes Full Marks: 30

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Course Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions	
CLO1	Explain the theories and principles of modern management and apply the concepts to the management of organizations in private and public sector
CLO2	Understand how managers can effectively plan in today's dynamic environment,
CLO3	Identify what strategies organizations might use to become more innovative and explain how the industrial company markets and price it's products and also how the company deal with it's environment.

Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**Ap**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	Managers have to engage in four basic managerial functions. Describe those functions and the linkage between them.	CLO1	U	5
1)	b)	Regardless of their level, managers may work in various areas within an organization. State at least 5 such common areas with brief descriptions.	CLO1	An	5
2)	a)	What are the 5 major competitive forces that affect organization-environment relationships? Describe those factors.	CLO2	R	5
2)	b)	Discuss about the task environmental elements. How do they differ from the general environmental elements?	CLO2	Ap	5
3)	a)	What do you mean by Job Specialization? What are it's advantages and disadvantages?	CLO3	U	5
3)	b)	What type of departmentalization you observed in IIUC? Describe its characteristics.	CLO3	E	5
		Or			
3)	a)	What do you mean by 'Authority'? How it is delegated to the subordinates?	CLO3	U	5
3)	b)	In your opinion, how important is it to have a clear and unambiguous chain of command? Why?	CLO3	E	5

International Islamic University Chittagong

Dept. of Computer Science & Engineering (CSE)

B.Sc. in CSE, Mid-Term Examination, Autumn 2025

Course Code: CSE 4871 **Title:** Neural Network and Fuzzy System

Total Marks: 30 Time: 1 hour 30 minutes

			CO	DL
1(a)	Discuss the role of Artificial Neural Networks as the core component of various learning algorithms in artificial intelligence.	3	CO1	C2
1(b)	What are activation functions? Give three examples with necessary graphical representation.	3	CO1	C2
1(c)	Draw the general model of Neural Network and explain it's processing steps. OR "A neuron can be defined with weighted average of input proceeding with activation function" – do you agree? Justify the answer with proper mathematical model.	4	CO1	C2
2(a)	What are the main applications of a Multilayer Perceptron (MLP), and how does it function? Explain in detail with an example and proper justification.	4	CO2	C3
2(b)	"Do you agree that backpropagation is a commonly used method for training neural networks? Why or why not? *OR* Neural network viewed as a directed graph-Explain the statement with a proper example.	3	CO2	C3
2(c)	Do you believe that the performance of a neural network is influenced by dropout? Explain your reasoning.	3	CO1	C2
3(a)	What is supervised machine learning and how does it relate to unsupervised machine learning?	3	CO1	C2
3(b)	With proper example explain Basic Concept of Competitive Learning Rule. *OR* Briefly explain algorithmic steps for K means Clustering.	3	CO2	C2
3(c)	With necessary graphical representation, design a simple neural network to classify handwritten digits.	4	CO2	C2

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Mid-term Examination, Autumn 2025

Course Code: CSE 4743 **Course Title: Computer Security**

Time: 1 hours 30 minutes *Total marks: 30*

[Answer the following questions. Figures in the right-hand margin indicate full marks.]

#Q	Questions	M	CLO	DL
Q1	a) The CIA triad (Confidentiality, Integrity, Availability) is foundational in security. Analyze how a Distributed Denial of Service (DDoS) attack violates this triad. Which element is most affected and why?	5	1	Ev
	b) Describe any three security mechanisms that are included in ITU X.800 recommendation.	5	1	U
Q1 Or	a) Compare active attacks and passive attacks in terms of detectability, objectives, and potential damage. In which real-world scenario would a passive attack be more dangerous than an active one?	5	1	Ev
	b) An attacker intercepts a fund transfer request in an online banking system and replays it multiple times to steal money. What type of security attack is this? Which security service(s) should the bank implement to prevent this in the future?	5	1	An
Q2	a) Analyze the differences between cryptanalysis of Transposition cipher and Caesar cipher.	5	1	An
	b) Given the following conditions for being a ring, show that the set of integers is a ring.	5	1	An

Ring	Abelian group	Group	(A1) Closure under addition:	If a and b belong to S, then $a + b$ is also in S
			(A2) Associativity of addition:	$a + (b + c) = (a + b) + c$ for all a, b, c in S
			(A3) Additive identity:	There is an element 0 in R such that $a + 0 = 0 + a = a$ for all a in S
			(A4) Additive inverse:	For each a in S there is an element $-a$ in S such that $a + (-a) = (-a) + a = 0$
			(A5) Commutativity of addition:	$a + b = b + a$ for all a, b in S
			(M1) Closure under multiplication:	If a and b belong to S, then ab is also in S
			(M2) Associativity of multiplication:	$a(bc) = (ab)c$ for all a, b, c in S
			(M3) Distributive laws:	$a(b + c) = ab + ac$ for all a, b, c in S $(a + b)c = ac + bc$ for all a, b, c in S

#Q	Questions	M	CLO	DL
Q3	a) In a **hybrid encryption model**, both link and end-to-end encryption are deployed. Analyze how this improves overall security. What is the main drawback of this approach?	5	3	U
	b) Write down 3 polynomials in GF (2^8). Show multiplication and addition in GF(2^8).	5	2	Ap p